

Varun Gupta

London, UK | MSc Artificial Intelligence (2026) | Graduate AI Engineer
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Right to work in the UK (Graduate Route eligible) | No sponsorship required

PROFILE

Graduate AI Engineer (MSc Artificial Intelligence, University of East London, 2026) who builds and ships LLM-powered and machine learning systems in Python. Built and **deployed** a live Retrieval-Augmented Generation application (embeddings, FAISS vector store, cosine-similarity retrieval, LLM generation) and multiple agentic-AI workflows. Hands-on across the AI build cycle: RAG, prompt engineering, NLP, model evaluation and testing, and deployment with Docker fundamentals, MLflow, and Git.

SKILLS

- Languages & Data:** Python, SQL, C++, Pandas, NumPy, data preprocessing, feature engineering
- LLMs & Generative AI:** RAG pipelines, embeddings, vector databases (FAISS), prompt engineering, agentic workflows, structured outputs
- Machine Learning:** scikit-learn, classification, regression, model evaluation & testing, threshold optimisation, imbalanced data (SMOTE)
- NLP:** TF-IDF, text classification, document Q&A, support-ticket automation
- MLOps & Deployment:** Docker fundamentals, Streamlit / Streamlit Cloud, MLflow, Git/GitHub, CI/CD basics

PROJECTS

- RAG Document Q&A Assistant** | *Python, RAG, FAISS, Embeddings, LLM, Streamlit (live)* 2026
 - Built and deployed a live RAG application serving grounded answers over user-uploaded PDFs by wiring a 6-stage pipeline (chunk, embed, index, retrieve top-k, prompt, generate) directly in Python without a framework.
 - Cut hallucination risk by returning the top-4 retrieved chunks with cosine-similarity scores alongside every answer, making 100% of responses traceable to source passages.
 - Reduced retrieval noise by tuning chunk size and overlap and normalising embeddings for cosine search via a FAISS inner-product index across all document chunks.
- AI Code Reviewer** | *Python, Agentic AI, Prompt Engineering, Structured Outputs* 2026
 - Built a multi-agent code-review pipeline coordinating 3 specialist agents (logic, code-quality, improvement) by decomposing review into separate prompt-engineered roles.
 - Standardised outputs across 3 review categories (bug, maintainability, design-quality) by enforcing structured JSON schemas for every agent response.
 - Improved repeatability of AI recommendations by separating review stages into documented, independently testable modules.
- Credit Card Fraud Detection** | *Python, scikit-learn, Model Evaluation* 2026
 - Improved fraud recall from 76% to 90%+ on 284,807 imbalanced transactions by applying SMOTE oversampling, stratified splitting, and decision-threshold optimisation.
 - Built a reproducible ML pipeline by modularising preprocessing, training, and evaluation into discrete Python stages.
 - Validated rare-event performance using precision-recall analysis rather than accuracy to handle the 0.17% fraud class imbalance.
- Customer Ticket Classifier** | *Python, NLP, TF-IDF, scikit-learn, Streamlit* 2026
 - Built an NLP classification pipeline by preprocessing unstructured text, applying TF-IDF vectorisation, and training supervised models with evaluation reporting.
 - Enabled both single-ticket and batch-CSV prediction by integrating model inference into an interactive Streamlit dashboard.
 - Improved input reliability by standardising a repeatable text preprocessing and feature-extraction stage ahead of inference.

EXPERIENCE

- IBOTIX Technologies** Aug 2023 – Sep 2023 India
 - RPA Intern – Automation & Data Processing*
 - Reduced manual processing effort by 50% on workflows handling 3,000+ records/day by automating repetitive data-processing tasks.
 - Reduced data errors and reprocessing by 30% across 5+ workflows by implementing validation checks, structured logging, retry logic, and exception handling.
 - Improved pipeline reliability by structuring ETL extraction, transformation, and validation steps following enterprise automation patterns.

EDUCATION

- University of East London** May 2025 – May 2026 London, UK
 - MSc Artificial Intelligence – Completed May 2026; passed all examinations*
- Jagannath International Management School (JIMS), Vasant Kunj** Dec 2021 – Jul 2024 India
 - Bachelor of Computer Applications – affiliated to GGSIP University*

CERTIFICATIONS

- Databricks Accredited Generative AI Fundamentals
- AWS — Introduction to Machine Learning; ML Terminology & Process
- MATLAB Onramps — Machine Learning, Deep Learning, Computer Vision, Image Processing